Mechanics Final Project: Chapter 15 Special Relativity

May 4, 2020

1 Background

Relativity is a branch of physics which can be used to describe the mechanics of physical systems in which particles in motion approach the speed of light. The name "relativity" refers to the manner in which measurements of physical processes are made relative to a chosen reference frame with a chosen origin in space and time.

Transformations can be made between measured properties in one frame and their observed values in another using what is called a Galilean Transformation. Suppose we have two reference frames $\mathcal{S}$ and $\mathcal{S}'$ with coordinates (x, y, z) and (x', y', z') respectively. $\mathcal{S}'$ is moving with a velocity V relative to $\mathcal{S}$. At time t, if light is sent up a distance h and reflected back to its starting point, the time that it returns as measured in $\mathcal{S}'$ could be described as $\Delta t' = \frac{2h}{c}$ where c is the speed of light. In frame $\mathcal{S}$, which the frame $\mathcal{S}$ is moving past, the light is following a more triangular path because the frame is moving in the horizontal direction while the light is moving up then down in that frame. Both paths can be seen in figure 1. The two right triangles formed will have a base length $\frac{V\Delta t}{2}$ with height h and we can measure the hypotenuse to be $\frac{c\Delta t}{2}$. Then using trigonometry, we find that, $(c\Delta t/2)^2 = h^2 + (V\Delta t/2)^2$. This allows us to find that $\Delta t = \frac{2h}{c\sqrt{1-\frac{V^2}{c^2}}}$. In relativity, we rewrite $\beta = \frac{V}{c}$, so this relationship becomes $\Delta t = \frac{\Delta t'}{\sqrt{1-\beta^2}}$. This relationship indicates an important physical law which is that $V < c$ so the speed of one inertial frame relative to another can never be equal to or greater than the speed of light.

The factor $\frac{1}{\sqrt{1-\beta^2}}$ is typically defined as γ and used often in relativity to convert measured values between frames and allows us to perform what are called Lorentz transformations to transfer between coordinate systems. Of particular interest to this problem is the relativistic momentum of a body. Using the classical relation $p = mv$ would cause momentum to not be conserved in all frames as the velocity of all bodies relative to an observer in each frame would vary. Thus, we must use the formula $p = \gamma mv$ in order to calculate a relativistic momentum which allows for that conservation of momentum to occur in all frames. We will be using the relativistic momentum to solve for the equations of motion of relativistic oscillatory motion in the following problem.

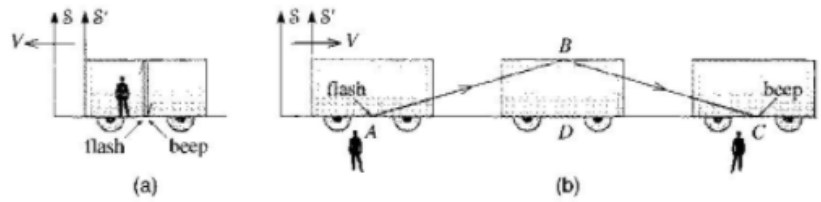

Figure 1: A diagram of a flash of light being sent up, then reflected back one in one reference frame moving relative to another. In this case, the reference frame $\mathcal{S}$ is a moving train car . In the car's frame (a), the light appears to move up to the top in a straight line, then be reflected back down along the same path. For an observer watching the train car go by in frame $\mathcal{S}'$, the path of this light appears to follow a triangular path as it gains a horizontal component from the velocity of the train(b). The beep is used to signify when the light has reached its final destination (Taylor, 604).

2 The Problem: Relativistic Oscillation

Consider a spring of constant k stretched to a length such that the attached particle approaches the speed of light as it oscillates. We are going to solve for the motion of this particle using a relativistic Lagrangian. In the case of this particle, there is no velocity dependence in the potential energy, so the relativistic Lagrangian can be written out as the ordinatry Lagrangian

$$\mathcal{L} = T - U$$

where T is the kinetic energy and U is the potential energy. The kinetic energy will vary from classical mechanics due to the relativistic nature of particle motion. Our first step to this problem is to find T. We can do this by initially setting our potential equal to 0, thus $\mathcal{L} = T$. We can then use the relativistic momentum to solve for the kinetic energy. The momentum can be written as

$$p_x = \frac{d\mathcal{L}}{d\dot{x}}.$$

Relativistic momentum is defined as

$$p_x = \gamma m \dot{x}$$

where the Lorentz transformation $\gamma = \frac{1}{\sqrt{1-\frac{\dot{x}^2}{c^2}}}$ and $\dot{x}$ is the speed of the object and c is the speed of light. Using the relationship between the relativistic momentum and $\frac{d\mathcal{L}}{d\dot{x}}$, we can multiply both sides by $d\dot{x}$ and then integrate:

$$\int p_x d\dot{x} = \int \frac{d\mathcal{L}}{d\dot{x}} = \int \gamma m \dot{x}.$$

We can solve this integral in Python as seen in figure 2 to get

$$\mathcal{L} = \frac{m\dot{x}^2}{2\sqrt{1-\frac{\dot{x}^2}{c^2}}} = T.$$

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In [43]:
t,k,m,c,v=sym.symbols('t,k,m,c,v')
x=sym.Function('x')(t)
xdot=x.diff(t)
xdouble=xdot.diff(t)

In [44]:
γ=1/sym.sqrt(1-xdot**2/c**2)
p=γ*(m*xdot)
sym.integrate(p,xdot)
```

Out[44]:

$$\frac{m\dot{x}^2}{2\sqrt{1-\frac{\dot{x}^2}{c^2}}}$$

Figure 2: Using sympy to solve the integral of the momentum with respect to $\dot{x}$ to find the Lagrangian $\mathcal{L}$ in Python.

```
In [45]:
T=m*xdot**2/(2*sym.sqrt(1-xdot**2/c**2)) #relativistic kinetic energy
U=(k*x**2)/2 #spring potential
L=T-U #Lagrangian

In [46]:
dLdx=L.diff(x) #derivative of the Lagrangian w.r.t. x
dLdxdot=L.diff(xdot) #derivative of the Lagrangian w.r.t x double dot
dLdt=dLdxdot.diff(t) #derivative of the above expression w.r.t. time
Lagrange=dLdx-dLdt #equation used to solve for motion
Lagrange
```

Out[46]:

$$-kx-\frac{m\ddot{x}}{\sqrt{1-\frac{\dot{x}^2}{c^2}}}-\frac{5m\dot{x}^2\ddot{x}}{2c^2\left(1-\frac{\dot{x}^2}{c^2}\right)^{\frac{3}{2}}}-\frac{3m\dot{x}^4\ddot{x}}{2c^4\left(1-\frac{\dot{x}^2}{c^2}\right)^{\frac{5}{2}}}$$

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In [47]:
x2dot=sym.solve(Lagrange,xdouble) #solving for the equation of motion
x2dot #x double dot
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Out[47]:

$$\left[-\frac{2k\sqrt{\frac{c^2-\dot{x}^2}{c^2}}\left(c^4-2c^2\dot{x}^2+\dot{x}^4\right)x}{c^2m\left(2c^2+\dot{x}^2\right)}\right]$$

Figure 3: After constructing the Lagrangian, we were able to solve for our equation of motion in the x direction using the formula $\frac{d\mathcal{L}}{dx}=\frac{d}{dt}\frac{d\mathcal{L}}{d\ddot{x}}$, from this, we generated the final output which is the expression for $\ddot{x}$

We can conclude that this Lagrangian is equal to our expression for the kinetic energy T.

In order to find the equations of motion, we next need to account for the potential energy in the Lagrangian expression. Since the potential of a spring is not dependent on velocity, we can just use the ordinary spring potential $\frac{1}{2}kx^2$. From here, we use the Lagrangian to solve for the equation of motion using the relation

$$\frac{d\mathcal{L}}{dx}=\frac{d}{dt}\frac{d\mathcal{L}}{d\ddot{x}}$$

Taking these derivatives, we were then able to solve for $\ddot{x}$ which shows us the acceleration of the particle at the end of the spring in the x direction, we solved for $\ddot{x}$ using the code found in figure 3. From this, we found that

$$\ddot{x}=\frac{-2k\sqrt{\frac{c^2-\dot{x}^2}{c^2}}(c^4-2c^2\dot{x}^4)x}{c^2m(2c^2+\dot{x}^2)}.$$

At slow speeds, this equation would reduce to the standard equation of motion for simple harmonic oscillators. In general relativity, we define the speed of light $c = 1$ and all other speeds are expressed as fractions of the speed of light. At slow speeds relative to the speed of light we can approximate the velocity $\dot{x} = 0$. Substituting this into our acceleration $\ddot{x}$ formula, we find that the equation reduces to $-\frac{kx}{m} = \omega^2 x$ where ω is the frequency of the oscillator. This is the standard equation of motion for simple harmonic motion. This is important to note because Newtonian mechanics is an appropriate approximation for "relativistic mechanics" at slower speeds. Relativistic transformations account for the shortcoming of classical mechanics in the case of objects approaching the speed of light. Otherwise, classical mechanics are sufficient for describing physical processes.

Next, we can solve for the motion of the particle over time. First, using Python, we substituted in the parameters $k = m = c = 1$ in order to solve for our acceleration, see figure 4 to see the code used and the calculated result. Using this result, we can then numerically

```
In [28]:  x2dot[0].subs({k:1,c:1,m:1})
```

Out[28]: $$-\frac{2\sqrt{1-\dot{x}^2}\left(\dot{x}^4 - 2\dot{x}^2 + 1\right)x}{\dot{x}^2 + 2}$$

Figure 4: Solving for the acceleration of the particle using the parameters $k = m = c = 1$.

solve for the motion of the particle. We can do so using a while loop where a list of the positions and velocities are generated. The code for this loop is in figure 5. Using the list, we can then plot the motion of the particle. Here, we do this for particles with different starting positions of $x_0 = 0.5, 1, 2, 4$. For each initial position, we re-initialize the same loop beginning at that x-position and generate a seperate list. The plot of the motions of a particle attached to a spring with the varying starting positions is seen in figure 6. In this plot of motion, we can see that when the spring is stretched to shorter lengths, the oscillatory motion looks fairly similar to an ordinary spring described by Newtonian mechanics, this is due to the fact that the motion is not allowed to accelerate to as high of a speed. At higher initial lengths, the motion jumps back and forth much more sharply We can also show how the velocity of the particle changes over time by plotting the values generated for velocity from

```
In [29]:
vx=0 #initial velocity of the partical
#for xo=0.5 being the initial position
x=0.5
t=0
dt=0.01
#empty lists to generate with the while loop
xs=[] #position to update
vxs=[] #velocity to update
ts=[] #time to update
while t<20:
    ax=-2*np.sqrt(1-vx**2)*(vx**4-2*vx**2+1)*x/(vx**2+2) #acceleration calculated using the Lagrangian
    vx=vx+ax*dt
    x=x+vx*dt
    xs.append(x)
    vxs.append(vx)
    t=t+dt
    ts.append(t)
```

Figure 5: While loop used to numerically solve for the motion of the particle. We begin with the acceleration and use this to update a list of velocities, then use the velocity to update the position over time. This particular loop generated a list of positions and velocities for the particle with an initial position of $x_0 = 0.5$

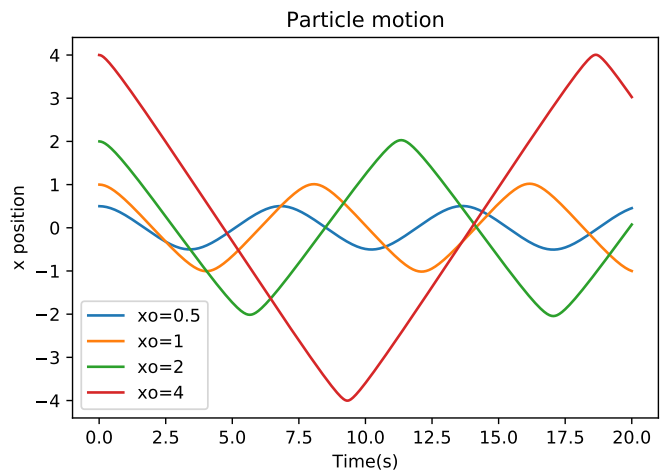

Figure 6: Position of the particle attached to the spring stretched sufficiently to allow the particle to approach the speed of light with initial positions $x_0 = 0.5, 1, 2, 4$.

our while loop as shown in figure 7. Here, we see again that at smaller initial positions, the spring follows a fairly classical oscillation pattern and as the spring begins at higher initial positions, the speed increases more due to having a greater amount of time to acclerate and experiencing a greater spring potential from the beginning. We can also see that the speeds never exceed $\dot{x} = 1 = c$ as no velocity can equal or exceed the speed of light. For initial lengths $x_0 = 2$ and $x_0 = 4$, both have an asymptotic flattening of their curves as the velocity approaches 1 indicating this physical limit. The plotted motion of the position and velocity of a spring stretched sufficiently to approach the speed of light demonstrates these properties of relativistic motion.

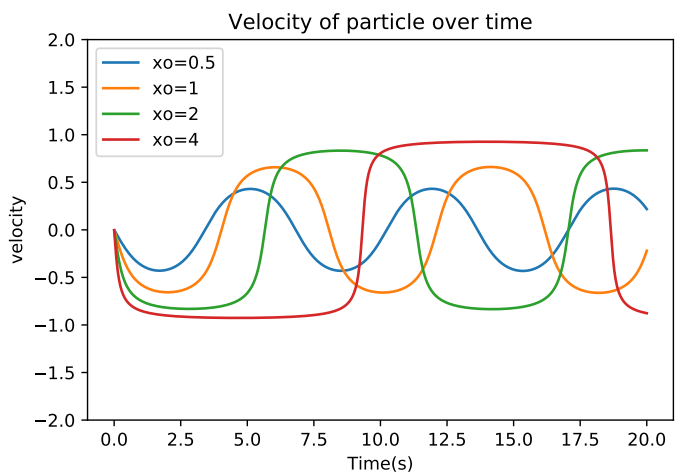

Figure 7: The velocity of a particle attached to a spring sufficiently stretched to cause the velocity of the particle to approach the speed of light with initial positions $x_0 = 0.5, 1, 2, 4$.